MKG APPLIED TO SUPPLY CHAIN INDUSTRY · FEBRUARY 2026

The supply chain sees *disruptions coming.* Too late, always.

Supply chain disruptions cost companies an average of 8% of annual revenue and occur every 3.7 years on average — each lasting over a

exist. They do. The problem is that no system reads them across the full relationship network before the disruption reaches your tier-1

THE SCALE OF THE GAP — 2025 DATA

8%
AVERAGE REVENUE LOST PER DISRUPTION EVENT · 2024

3.7yr
AVERAGE TIME BETWEEN MAJOR DISRUPTIONS LASTING >1

94%
COMPANIES REPORT REVENUE IMPACT FROM DISRUPTIONS
Procurement Tactics 2025

6%
COMPANIES WITH FULL SUPPLY CHAIN VISIBILITY — ONLY 6%
Gartner 2025

40%+
ORGANIZATIONS WITH LIMITED OR NO TIER 1 SUPPLIER VISIBILITY

45%
OF A DECADE'S PROFITS CAN BE LOST TO DISRUPTIONS (MCKINSEY)

suppliers. MKG changes that.

McKinsey / WEF 2025

McKinsey Global Institute

SECTION 01 · THE VISIBILITY PROBLEM

Three-tier visibility. *Only one tier is actually visible.*

The exact same structural gap that exists in financial markets — Speed 1 vs Speed 2 vs Speed 3 — exists in supply chains. Tier-1 suppliers are well-monitored. Tier-2 suppliers are barely visible. Tier-3 and beyond are completely dark. But disruptions almost always originate there.

TIER	WHAT IT COVERS	VISIBILITY & TOOLS
TIER 1 DIRECT SUPPLIERS	**Your direct, named suppliers — the ones on your contracts** Companies you buy directly from. Performance monitored via SLAs, QBRs, ERP data, supplier portals. Financial health tracked via D&B, Moody's. Lead times measured. Usually 50–200 suppliers. Well-understood but still incomplete — 40%+ of companies lack even full Tier 1 visibility. ▸ Example: Apple → TSMC (direct wafer order), Toyota → Denso (direct parts), Nike → Yue Yuen (direct manufacturing) ▸ Disruption warning time: Days to weeks — usually caught via supplier portal, financial alerts, news	WELL MONITORED ERP / SRM tools SAP Ariba, Coupa D&B / Moody's alerts Resilinc tier-1

TIER 2 SUB-SUPPLIERS	**Your suppliers' suppliers — the companies you don't contract with but depend on** Companies that supply materials, components, or services to your Tier 1 suppliers. Rarely in your ERP. Often unknown to you by name. Only 43% of companies have visibility into Tier 2 at all. Yet Tier 2 disruptions cascade into Tier 1 failures 70–80% of the time. The most undermonitored, highest-impact tier. ▸ Example: TSMC → ASML (EUV machines), Denso → Renesas (chips), Yue Yuen → fabric mills (raw materials) ▸ Disruption warning time: Weeks to months — usually discovered only AFTER the Tier 1 begins failing	PARTIALLY VISIBLE Interos (partial) Resilinc (partial) 43% of firms only
TIER 3+ DEEP CHAIN	**Sub-sub-suppliers and beyond — the true origin of most major disruptions** Raw material producers, specialized chemical suppliers, rare earth miners, single-source component makers. Almost completely invisible to most companies. COVID-19 exposed how deeply companies depend on Tier 3+ suppliers they didn't even know they had. This is where the 2021 automotive chip shortage actually originated — Tier 3 chemical plant fire in Japan. ▸ Example: Renesas ← Shin-Etsu Chemical (silicon wafers) ← quartz mining operations in Brazil — 3 hops from Toyota ▸ Disruption warning time: Months — by the time you know, your Tier 1 is already shutting lines	INVISIBLE No tool covers this 100% dark MKG target zone

"About half of a disruption's total effect may come from amplification through the supply chain network. A quarter of GDP and inflation effects after 2020 are attributed to shocks outside the U.S. that propagated through input-output networks."

FEDERAL RESERVE BANK OF RICHMOND · ECONOMIC BRIEF · JANUARY 2025

SECTION 02 · DISRUPTION TAXONOMY

Six types of supply chain shock. *All transmit through the graph.*

Every major supply chain disruption follows the same pattern: a signal appears in the news — a typhoon path, a factory fire, a bankruptcy filing, a regulatory change — days or weeks before the disruption reaches your supply chain. MKG reads those signals in real time across all six disruption types.

DISRUPTION TYPE 01

Geopolitical Shock

Trade wars, sanctions, tariffs, export controls, port closures due to conflict. Red Sea crisis (2023–24): $6B/week in trade disrupted. Russia-Ukraine: 35% decline in Ukrainian agricultural output. US-China tariffs: 17% drop in Chinese imports to US in 2025.

38% rise in disruptions from geopolitical causes (2024 vs 2023)

⏱ Signal-to-impact window: 2–8 weeks · Detectable in news 14–30 days before supply chain effect

News detectable

Graph-traversable

Tier 3 originating

DISRUPTION TYPE 02

Climate & Weather Event

Hurricanes, droughts, flooding, wildfires. Panama Canal drought: 32% reduction in transit capacity, 21-day delays. Hurricane Helene (2024): semiconductor-grade quartz supply impacted. 41% of organisations investing in weather resilience (2024).

Aerospace & defense: $184M/yr disruption cost (Gartner 2025)

⏱ Signal-to-impact window: 5–21 days · Weather tracking detectable 7–14 days in advance

Predictable signals

Route graph mapping

Cascades deep

DISRUPTION TYPE 03

Cyber & Technology Failure

Third-party software attacks, ERP failures, ransomware. Jaguar Land Rover: £72M/day production halt from cyber attack. 35.5% of all data breaches involve third parties (2024, up from 29%). Software supply chain attacks +68% in 2025. Industrial breach avg: $5.56M.

$138B predicted annual cost of SC cyber attacks by 2031

⏱ Signal-to-impact window: Hours to days · Software vendor alerts often precede cascades

Fast-moving

Vendor graph critical

Shared software nodes

DISRUPTION TYPE 04

Financial Instability

Supplier bankruptcies, credit stress, force majeure declarations. Bankruptcies up 200% in H1 2024 vs H1 2023. Force majeures up 128%. Hanjin Shipping

DISRUPTION TYPE 05

Regulatory & Trade Change

Export controls, compliance mandates, tariffs, environmental regulations. EU CSDD, UK Modern Slavery Act, US CHIPS Act, SEC

DISRUPTION TYPE 06

Operational Failure

Factory fires, equipment failures, labor strikes, leadership transitions. Factory fires down 21% in 2024 but still major cause. Labor unrest: 3,000+ strikes

bankruptcy (2016): billions of cargo stranded. Financial distress signals appear in filings 60–90 days before collapse.

200% increase in supplier bankruptcies · H1 2024 vs H1 2023

⏱️ Signal-to-impact window: 60–120 days · Detectable via financial news and filing monitoring

Long signal window

Filing-detectable

Cascades to tier-2/3

climate disclosure. Regulatory change disruptions up 185% in H1 2024. China REE export restrictions on yttrium (April 2025): immediate semiconductor cascade.

185% increase in regulatory-driven disruptions · H1 2024

⏱️ Signal-to-impact window: Weeks to months · Policy text published before effective date

Long lead time

Regulatory entity graph

~Partially covered

tracked globally. Leadership transitions up 92% in H1 2024 — each creates supply chain uncertainty. Single-source component failures.

Leadership transitions up 92% · H1 2024 vs H1 2023 · High single-source risk

⏱️ Signal-to-impact window: Days to weeks · Factory incident news precedes supply impact by 5–14 days

News-detectable

Concentrated risk

Single-source mapping

SECTION 03 · LIVE USE CASE SCENARIOS

Real disruptions. Real signals. *How MKG would have seen them first.*

Three of 2024–2025's most damaging supply chain events, re-examined through MKG's graph architecture. In each case, the signals existed days to weeks before the impact. No company had a system to connect them.

SCENARIO 01 · GEOPOLITICAL / MARITIME

Red Sea Crisis — Houthi Attacks on Shipping Lanes · Late 2023 → 2024

$6B/week trade disrupted · 35% increase in lead times · Shipping costs +15-20% · 30,000+ vessels rerouted

WITHOUT MKG — WHAT COMPANIES EXPERIENCED

T-30 DAYS

Houthi attacks on shipping begin. International news.

News reported. Most supply chain teams monitoring but no systematic mapping of which of their Tier-2/3 suppliers' components transit the Red Sea regularly.

T-14 DAYS

WITH MKG — GRAPH-DRIVEN EARLY WARNING

T-45 DAYS — SIGNAL DETECTED

MKG TRIGGER

Escalation news in regional Arabic and English press.

WAN reads news signals: Houthi escalation + shipping lane threat. Graph traversal begins: which of your suppliers (all tiers) source components routed via Red Sea? Scans vessel routing data + supplier

Major shipping lines announce Red Sea avoidance.

Maersk, MSC, CMA CGM divert vessels. Most companies learn this from general logistics news, not from a supplier-specific alert. No automated mapping of which specific purchase orders are on affected vessels.

T-3 DAYS

Procurement begins frantic supplier calls.

Teams manually calling Tier-1 suppliers to identify exposure. No Tier-2/3 mapping. Partial picture only. Spot market scramble begins — at 200–400% premium freight rates.

T+0 WEEKS

Production lines begin slowing. Inventory runs out.

First material shortages hit assembly lines. Companies discover they had Tier-3 sub-suppliers in Asia whose sole routing was via the Red Sea — completely unknown before the crisis.

T+4 WEEKS

Full impact: 35% lead time increase. Avg 10–14 day delays.

Total exposure finally understood. Far too late to mitigate. Stock procurement costs surge. Customer SLA breaches begin. Force majeures triggered downstream.

geography edges. Alert generated in 60 seconds.

T-38 DAYS

MKG OUTPUT

Ranked exposure list delivered to CPO dashboard.

Impact score per supplier (all tiers). Red: 12 Tier-2/3 suppliers with >60% Red Sea dependency. Amber: 8 Tier-1 suppliers with partial exposure. Green: 23 suppliers re-routable via Cape. Alternative routing options pre-computed. Freight cost delta modelled.

T-30 DAYS

Procurement begins targeted mitigation.

High-exposure suppliers contacted directly. Air freight pre-booked at pre-crisis rates for critical components. Buffer stock orders placed. 3 weeks before the market scramble — at 60–70% cost saving vs. spot.

T+0

Disruption hits — company is pre-positioned.

Most critical components already in warehouse. Spot freight avoided. Competitor lines slowing while your lines continue. Customer commitments maintained. Insurance claim documentation automatically generated from MKG's graph trace.

VALUE GENERATED

38 days of early warning vs. 3 days. Pre-booked freight at pre-crisis rates. Tier-3 exposure mapped automatically — which took competitors 4+ weeks of manual research. Force majeures avoided. Customer SLAs maintained. Conservative estimate: $2–15M in avoided costs per major disruption event for a Global 500 manufacturer.

SCENARIO 02 · SINGLE-SOURCE / OPERATIONAL FAILURE

Renesas Factory Fire → Global Automotive Chip Shortage · March 2021 (Repeat Pattern 2024)

Single Tier-3 plant fire → Ford, GM, Toyota production halts · $210B in automotive revenue lost globally · Months to recover

THE HIDDEN CHAIN THAT DESTROYED PRODUCTION

The 2021 Renesas factory fire is the canonical Tier-3 blind spot case. Renesas supplied automotive MCUs. But automotive OEMs didn't buy directly from Renesas — they bought from Tier-1 suppliers (Bosch, Denso, Continental) who incorporated Renesas chips. The chain was:

```
Toyota / GM / Ford
↓ buys from (Tier 1)
Bosch / Denso / Continental
↓ buys from (Tier 2)
Renesas Electronics
↓ buys from (Tier 3)
Shin-Etsu Chemical (silicon wafers)
🔥 Fire at Naka fab → entire chain collapses
```

No automotive OEM had Renesas mapped as a critical Tier-3 dependency. None had Shin-Etsu in their supply chain monitoring system at all. The signal — a fire at a single Japanese factory — was visible globally within hours. But no system connected that fire to Toyota's production lines 3 hops away.

> THE TRIBAL KNOWLEDGE THAT WAS LOST
>
> The one person at Ford who knew Renesas was a critical Tier-3 sub-

HOW MKG WOULD HAVE SEEN IT

T+0 HOURS — FIRE REPORTED

MKG TRIGGER

Factory fire news published globally.

WAN reads news: "Renesas Naka fab fire." Entity extraction: Renesas Electronics, Naka facility, Japan, semiconductor fabrication. Graph traversal begins immediately across all supplier edges where Renesas appears — including Tier-2 and Tier-3 edges.

T+60 SECONDS

MKG OUTPUT

Full 3-hop impact assessment delivered.

Alert delivered to CPO dashboard: "Renesas Naka fire affects 47 entities in your supply network across 3 tiers. Criticality score 91/100 (single-source, no substitution). Tier-1 affected: Bosch (4 product lines), Denso (3 product lines). Production impact estimate: 14–21 day delay per platform."

T+6 HOURS

Competitor analysis begins automatically.

Graph traversal extended: which competitors use same Renesas MCUs? Texas Instruments and Infineon alternatives identified with current availability scores. Secondary suppliers who could ramp production scored by capacity and qualification status.

T+1 DAY

Procurement acts before market tightens.

Alternate source orders placed before industry-wide panic buying begins. 3 weeks before competitors know they have a problem. Price locks secured. The company that moved first captures remaining spot inventory at pre-shortage prices.

> supplier had left the company 18 months earlier. Their mental map of the dependency left with them. This is the tribal knowledge destruction problem — exactly what MKG's temporal graph prevents.

SCENARIO 03 · REGULATORY / GEOPOLITICAL · 2025

China Rare Earth Element Export Restrictions · April 2025

China restricts yttrium exports → catalytic converters, TV screens, semiconductor manufacturing → cascading global manufacturing impact

China controls 60–80% of rare earth mining and 85–90% of rare earth refining globally. When China imposed export restrictions on yttrium in April 2025, companies using yttrium-based materials — catalytic converters, semiconductor phosphors, TV displays, fuel cells — faced an immediate upstream supply shock.

The problem: most companies using yttrium don't buy it directly. They buy finished components from Tier-1 suppliers who incorporate yttrium-based materials from Chinese refiners they've never heard of. The regulatory change was published in Chinese government press releases — available in Chinese before English translation.

> THE LANGUAGE GAP
>
> The yttrium export restriction announcement appeared in Chinese-language regulatory publications 11 days before English-language trade

MKG REGULATORY GRAPH — HOW IT WORKS

MKG treats regulatory bodies as first-class graph nodes. China's Ministry of Commerce (MOFCOM) is a node. The export restriction is an edge event. All companies with yttrium-dependent supply chains are reachable from that node via the regulatory co-exposure graph.

```
MOFCOM Export Restriction (node)
→ governs (regulatory edge)
Chinese REE refiners (6 companies)
→ supplies (SC edge, w=0.94)
Yttrium phosphor manufacturers (12 co.)
→ supplies (SC edge, w=0.71)
Automotive/Semis/Display Tier-1 suppliers
→ supplies (SC edge)
YOUR COMPANY ← alert here, T+60 seconds
```

DAY 1 — CHINESE PRESS RELEASE

MKG READS IN CHINESE

MOFCOM restriction published. WAN triggers immediately.

11 days before English-language coverage. Graph traversal identifies all companies co-exposed through yttrium dependency. Impact ranked by yttrium content per unit, substitutability, inventory days on hand.

> press coverage. Companies without multilingual news monitoring missed 11 days of warning time. MKG's 13-language scraping agent has no language gap — Chinese regulatory filings are read in real time.

DAY 2

Procurement secures strategic stockpile ahead of the market.

Order placed with non-Chinese yttrium refiners (Australia, Canada, Brazil) before panic buying. Long-term supply contracts locked at pre-restriction prices. Engineering begins qualification of yttrium-reduced formulations.

SECTION 04 · SUPPLY CHAIN TRIBAL KNOWLEDGE

What a 20-year supply chain veteran *knows that no tool captures*

The same tribal knowledge problem that exists in financial markets lives inside every corporate supply chain function. Senior SCM leaders carry relationship maps, supplier calibrations, and disruption patterns in their heads. When they leave, so does the institutional memory — and the company's resilience.

01 Single-Source Knowledge

The Hidden Single-Source Dependencies Nobody Documented

Every experienced SCM leader knows 10–20 "silent single-source" dependencies — components that look like they have alternates in the system but in practice cannot be switched without 18 months of requalification. This knowledge is built from requalification failures, quality incidents, and institutional memory of past substitution attempts. It is never systematically captured in ERP or sourcing systems.

> *"We have three listed alternates for that gasket material. But anyone who's been here 5 years knows none of them qualify for the chemical resistance spec. We're effectively single-sourced and the system doesn't know it."*

MKG BRIDGE

MKG's graph encodes qualification constraints as edge metadata. "Alternate exists but unqualified" becomes a graph property — a distinct edge type from "certified alternate." Graph traversal scores single-source risk correctly, not based on what the system says exists but on what is actually substitutable. Survives leadership departure.

02 Supplier Calibration

Supplier-Specific Behavioral Calibration — "This one always overpromises recovery"

Experienced procurement managers develop precise, company-specific calibration for how suppliers communicate during disruptions. Some suppliers always understate severity to protect the relationship. Others are transparent and accurate. Some consistently recover 30% faster than their initial estimate. Others chronically miss recovery dates. This calibration is never in the supplier scorecard system — it lives in the relationship manager's judgment.

> *"When that supplier says '4-week recovery,' we plan for 8. Every time, for 12 years. It's not in their scorecard — it's just what everyone on the desk knows. When we get new account managers they always get burned by it the first time."*

MKG BRIDGE

Historical disruption event data per supplier: stated recovery date vs. actual recovery date across all recorded events. Systematic over/under-promise bias becomes a computable calibration factor per supplier entity. New procurement managers inherit the institutional calibration on day one — the pattern survives even complete team turnover.

03 Canary Suppliers

The Canary Suppliers — "This one always signals what's coming for the whole sector"

In every industry, there are small, under-monitored suppliers whose operational changes systematically lead the entire sector. A small specialty chemical company whose order book changes always precede semiconductor lead time shifts by 6 weeks. A specialized logistics provider in a single region whose capacity utilisation signals broader logistics tightening before it becomes visible in rates. Veterans know these canaries. They watch them obsessively. No ERP or SCRM system tracks them as leading indicators.

> *"There's a specialty substrate supplier in Taiwan — 8 companies in the world make what they make. When their lead times go from 4 weeks to 10 weeks, every automotive OEM's tier-1 chips follow within 6 weeks. I've seen it happen four times."*

MKG BRIDGE

Lead-lag edge detection: edges with consistent temporal precedence (Supplier A's signals precede industry-wide disruptions by N weeks, validated across historical events) are encoded as "canary" edge types with lead-time metadata. The canary relationships become graph queries, discoverable by any analyst — not just the veteran who spent years figuring it out.

04 Relationship Strength

Real Relationship Weight vs. Contracted Volume — What the Relationship Actually Is

In supply chain databases, relationships are binary (supplier or not) or measured by contracted volume. But the real weight of a relationship — the willingness to prioritise your order in a shortage, the goodwill built over 20 years of collaboration, the personal relationship between your CPO and their CEO — is nowhere in any system. During a shortage, this informal relationship strength determines whose orders get filled first. Veterans know exactly which suppliers will "take care of" them when capacity is constrained.

> *"On paper we're their 8th largest customer. In practice their CEO calls our CPO personally before allocating scarce capacity. That relationship is worth $50M in a shortage. It's not in any database."*

MKG BRIDGE

Multi-dimensional edge weights: contracted volume weight (from procurement systems) + relationship quality signals (from news co-mentions, press release joint references, executive relationship signals from LinkedIn/filings) + historical allocation behavior during past shortages. A composite relationship strength score that captures more than pure volume. Encoded, not just remembered.

05 Disruption Pattern Memory

Historical Disruption Pattern Library — "This exact pattern happened in 2011"

Experienced supply chain leaders carry a mental library of historical disruption patterns. They immediately recognize when a current situation matches a past pattern — and they know exactly how it played out: which suppliers failed first, which alternatives were available, how long recovery took, what the sequence of events was. This pattern recognition drives faster, more accurate response. It is entirely unencoded in any system and completely lost when leaders retire.

> *"This is almost identical to the Japan earthquake in 2011. Same semiconductor substrate issue, same supplier concentration. In 2011 it took 14 weeks to recover. The recovery sequence went: X first, then Y, then*

MKG BRIDGE

Temporal graph rewind: historical disruption events stored as graph state snapshots. "Compare current situation to March 2011 Japan earthquake event." MKG surfaces historical propagation sequence, recovery timeline, and which alternative sources activated in sequence. The institutional memory of every past disruption becomes a queryable baseline for current event response — available to the most junior analyst on their first day.

Z. Let me call our Y supplier now before they get overwhelmed."

SECTION 05 · SUPPLY CHAIN BUYER PROFILES

Who feels this pain most. *Precisely.*

Six distinct buyer profiles within the supply chain function — each with specific pain points that MKG's graph architecture directly addresses. Unlike financial markets where the primary buyer is the PM, supply chain has multiple decision makers who each face the same underlying problem from a different angle.

PRIMARY BUYER · C-SUITE

Chief Procurement Officer (CPO)

Accountable for supply continuity across the enterprise. Board-level pressure since COVID. "Supply chain resilience" now the #1 or #2 priority for 80%+ of CPOs (Hackett Group 2025). Personally blamed when production halts. Spends enormous budget on risk tools that only cover Tier 1 — while most disruptions originate in Tier 2/3.

MKG VALUE

Board-level supply chain risk visibility across all tiers. Automatic disruption early warning 14–45 days before production impact. Regulatory compliance documentation for EU CSDD automatically generated from graph trace. Institutional memory that survives team turnover.

PRIMARY BUYER · RISK FUNCTION

Head of Supply Chain Risk

Tasked with mapping and monitoring supplier ecosystem risk. Current tools: static supplier databases, periodic audits, D&B alerts for Tier-1. Cannot monitor Tier-2/3 systematically — too many suppliers, too few analysts. 80% of their time is reactive (responding to disruptions that have already hit) rather than proactive (detecting signals before impact).

MKG VALUE

Continuous, automated multi-tier monitoring across all six disruption types. Analyst time shifts from reactive firefighting to proactive signal review. Every disruption comes with a full causal chain — graph path from news signal to affected entities — for audit documentation.

PRIMARY BUYER · OPERATIONS

VP Supply Chain / COO

Responsible for production continuity. Sits between the CPO's risk strategy and the factory floor's operational reality. When a line goes down, this person is the one scrambling to find alternate sources, authorize premium freight, and explain to the CEO why they didn't see it coming. Needs disruption intelligence in operational, actionable form — not academic risk reports.

MKG VALUE

Operational-format alerts: ranked impact list with affected production platforms, estimated delay per line, pre-computed alternate sourcing options with current availability scores. From signal to recommended action in one dashboard view. Not

a risk report — an action plan.

SECONDARY BUYER · IT/SECURITY

CISO / Supply Chain Cybersecurity Lead

Responsible for third-party cyber risk across the supplier ecosystem. Cannot monitor 5,000+ suppliers manually. Needs automated detection of software supply chain vulnerabilities that could cascade through their vendor network. 35.5% of breaches now involve third parties — and the chain propagates exactly like a supply chain network.

MKG VALUE

Software vendor co-dependency edges in the graph: companies sharing the same critical software stack (ERP, MES, WMS providers) are connected by "cyber co-exposure" edges. A breach at a shared software vendor triggers automatic traversal to all co-exposed supplier nodes. The cyber attack vector becomes graphable and monitorable.

SECONDARY BUYER · LEGAL/COMPLIANCE

Chief Compliance Officer

Accountable for EU CSDD, UK Modern Slavery Act, US CHIPS Act, SEC climate disclosure — all of which require documented supply chain due diligence. Cannot do this without a multi-tier supplier map. Current process: annual surveys, periodic audits — static snapshots. Regulators increasingly expect continuous monitoring, not point-in-time documentation.

MKG VALUE

Regulatory compliance graph: regulatory body nodes + governed entity edges. Every new regulation maps automatically to all affected suppliers. Continuous monitoring generates timestamped compliance evidence. Audit documentation auto-generated from graph trace — no manual assembly. Temporal graph enables point-in-time historical proof for regulatory inquiries.

TERTIARY BUYER · FINANCE

CFO / Supply Chain Finance Lead

Responsible for working capital optimization, supply chain finance programs, and the financial risk of supplier concentrations. Current challenge: cannot see financial stress signals in Tier-2/3 suppliers before they cascade to production failures and balance sheet write-downs. Bankruptcies of key Tier-3 suppliers create inventory write-downs they never expected because the dependency was invisible.

MKG VALUE

Financial health signals mapped to supplier graph nodes — credit ratings, D&B signals, filing anomalies, news sentiment — continuously updated. When a Tier-3 supplier shows financial distress signals, graph traversal computes which production lines are at risk and the revenue exposure. The CFO sees financial supply chain risk before it becomes a P&L event.

SECTION 06 · WHAT EXISTING TOOLS DON'T COVER

The supply chain tool stack. *What's missing from every one of them.*

Supply chain teams spend $50K–$500K+ per year on SCRM tools. Here is exactly what those tools cover and where MKG's graph adds what none of them provide.

Capability	SAP Ariba	Resilinc	Interos	Z2Data	Coupa	D&B / Moody's	M
TIER VISIBILITY							
Tier 1 supplier database	✓	✓	✓	✓	✓	✓	
Tier 2 supplier visibility	✗	~	✓	~	✗	✗	
Tier 3+ deep chain mapping	✗	✗	~	✗	✗	✗	
Single-source concentration scoring	✗	~	✓	✓	✗	✗	
DISRUPTION INTELLIGENCE							
Real-time disruption news monitoring	✗	✓	✓	~	✗	~	
Multilingual monitoring (6+ languages)	✗	~	✗	✗	✗	✗	
Automated disruption → SC impact mapping	✗	~	✓	~	✗	✗	
Multi-hop contagion traversal (3+ tiers)	✗	✗	✗	✗	✗	✗	N
60-second signal-to-alert latency	✗	✗	✗	✗	✗	✗	N
GRAPH & RELATIONSHIP INTELLIGENCE							
Dynamic edge weights (auto-updated)	✗	✗	✗	✗	✗	✗	N
Canary supplier lead-lag detection	✗	✗	✗	✗	✗	✗	N
Relationship decay detection (auto)	✗	✗	✗	✗	✗	✗	N
Supplier calibration (recovery bias model)	✗	✗	✗	✗	✗	✗	N
Regulatory co-exposure graph	✗	✗	~	✗	✗	✗	N
TEMPORAL & HISTORICAL INTELLIGENCE							
Historical disruption pattern replay	✗	✗	✗	✗	✗	✗	N
Temporal graph state (full rewind)	✗	✗	✗	✗	✗	✗	N
Institutional tribal knowledge encoding	✗	✗	✗	✗	✗	✗	N
COMPLIANCE & EXPLAINABILITY							

Capability	SAP Ariba	Resilinc	Interos	Z2Data	Coupa	D&B / Moody's	M
EU CSDD compliance documentation	~	~	✓	~	✗	✗	
Auto-generated audit trail (graph trace)	✗	✗	✗	✗	✗	✗	N
On-premise / air-gapped deployment	~	✗	✗	✗	✗	✗	
FINANCIAL & ERP INTEGRATION							
ERP procurement workflow integration	✓	~	✗	✗	✓	✗	
Supplier financial health monitoring	✗	✗	✓	✗	✗	✓	
Revenue-at-risk quantification	✗	✗	✗	✗	✗	✗	N

✓ = Full capability | ~ = Partial | ✗ = Not available | ✦ NEW = First commercial encoding | Bold rows = MKG-exclusive

THE CORE INSIGHT — SUPPLY CHAIN EDITION

SAP Ariba tells you who your suppliers are. Resilinc tells you when a disruption has been reported. Interos scores your Tier-1 and Tier-2 risk. D&B monitors your suppliers' financial health.

But no tool in the market can answer: "A fire just hit a semiconductor substrate plant in Kyushu, Japan — which of my production lines are affected, in what sequence, with what expected delay, through which suppliers, with what alternates available — in the next 60 seconds?"

That answer has required a 20-year veteran to sit at a

desk and manually piece together a relationship map that exists only in their head. When they retire, the map goes with them. Every new disruption, you start over.

MKG encodes the map. Keeps it current. Reads signals in 13 languages the moment they appear. Traverses the network automatically. Delivers the answer in 60 seconds — to every analyst, on day one, regardless of experience.

The disruption doesn't change. The preparation does.

MKG · SUPPLY CHAIN APPLICATION ANALYSIS · FEBRUARY 2026 · CONFIDENTIAL

3 Live Scenarios · 6 Disruption Types · 5 Tribal Knowledge Types · 6 Buyer Profiles